ENGLISH PAPER SUMMARY | ALL ORIGINAL NUMBERED FIGURES

Winter Ice-Wave Modeling With WAVEWATCH III in Lake Erie

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In the tested Lake Erie configurations, IC4M4 improves winter wave prediction over binary ice blocking. Greater physical complexity does not guarantee greater skill, and the accuracy of the ice forcing remains a major limitation.

Prepared independently from the original paper. This is a new English synthesis, not a translation of the Korean notes. The figure atlas reproduces all 10 numbered figures with explanatory reading notes. Section and equation references refer to the source publication.

Research question and model architecture

The paper evaluates twelve WAVEWATCH III ice-damping configurations against winter observations. Its motivation is operational: the IC0 scheme treats sufficiently ice-covered cells as land, eliminating wave guidance where partial or thin ice can still transmit waves. An undamped case supplies an additional reference. [Sections 1 and 2.4]

FVCOM-CICE is run first, and hourly ice concentration is then passed to WW3 version 6.07.1. Wind and ice concentration force WW3; waves do not feed back to FVCOM-CICE. Thus this is a loosely, one-way coupled configuration, rather than a direct test of wave-driven fracture. The Lake Erie mesh has 11,509 elements and 6,106 nodes, with approximately 350 m nearshore to 2.5 km offshore spacing. [Sections 2.1-2.2]

Observations and fair comparisons

The main damping comparison uses winter 2010-2011 AWAC observations at LE4, LE5, and LE6. Additional testing uses LE21 in 2012-2013 and pseudo-real-time forecasts at Reno Beach in 2022-2023. SWIPS measurements assess ice presence and thickness. The warm 2011-2012 season is not used for the ice-focused analysis because little ice formed at the open-water stations. [Section 2.2]

Point ice-thickness observations and areal concentration fields are different observables. For a coarse presence/absence comparison, SWIPS ice thickness of at least 5 cm and modeled or analyzed concentration of at least 50% define ice presence. This is a diagnostic approximation, not a direct thickness validation of the concentration product. [Section 2.3]

The tested parameterizations include simple blocking, constant damping, viscoelastic models, and empirical IC4 relationships. Parameter choices follow the literature, selected tests, or representative Lake Erie conditions; the paper does not comprehensively optimize every scheme. Consequently, rankings apply to the configurations and observations examined, not to every possible calibration. [Section 2.4 and Table 2]

Quantitative evidence

At LE4 for 1 January-15 April 2011, IC4M4 gives RMSE 0.32 m and bias -0.06 m, versus IC0 at 0.46 m and -0.23 m. At LE5, the corresponding IC4M4/IC0 RMSEs are 0.38/0.59 m; at LE6 they are 0.39/0.54 m. IC4M4 has the smallest absolute bias among the damping cases at all three stations. [Section 3.2 and Table 4]

This is not a universal ranking of every model at every station. The undamped case has lower RMSE than IC4M4 at LE5 and LE6, and IC2 has the lowest damped-case RMSE at LE6. Sparse and event-biased observations, including measurement cutoffs, affect these comparisons. The time series provide context that a single score cannot capture. [Section 3.2; Figures 3-4]

Many frequency-dependent or viscoelastic configurations damp waves too strongly, while undamped simulations can create excessive wave events. IC4M4 generally retains small waves in partial ice and gives a more plausible response to wind forcing than the binary ice mask. Tests in 2012-2013 and the 2022-2023 forecast configuration support its advantage over IC0, while still showing missed or underpredicted events. [Figures 4-7 and 10]

Why the lake differs from polar calibration settings

Lake Erie waves are typically shorter-period than waves in the polar ocean datasets underpinning many schemes. The paper contrasts peak frequencies around 0.15-0.20 Hz in the lake with approximately 0.05-0.10 Hz in the Arctic. A frequency law calibrated for longer ocean waves can therefore produce excessive attenuation in lake conditions. Figures 8-9 illustrate the wave-height, peak-period, and spectral differences. [Discussion and Section 4]

IC4M4 is based on a wave-height relationship developed for Antarctic conditions. Its good performance here is empirical evidence for this setup, not proof that all Antarctic physics transfers to freshwater. The study does not demonstrate that a wave-height-only scheme resolves all spectral or ice-state dependence.

Remaining uncertainty and research implications

Ice formation and especially melting can respond too slowly in FVCOM-CICE. Both the ice model and the daily ice analysis can disagree with local sonar observations. Incorrect ice persistence contributes to overdamping independently of the selected wave sink. Clouds, spatial support, and limited winter observations complicate validation. [Sections 3.1-3.2; Figure 2]

The authors also identify winter wind-stress representation and absent wave-induced breakup as possible improvements. They do not quantify breakup's contribution in these experiments. A useful Great Lakes benchmark should therefore include IC0, IC4M4, an undamped reference, and explicit sensitivity to ice and wind forcing.

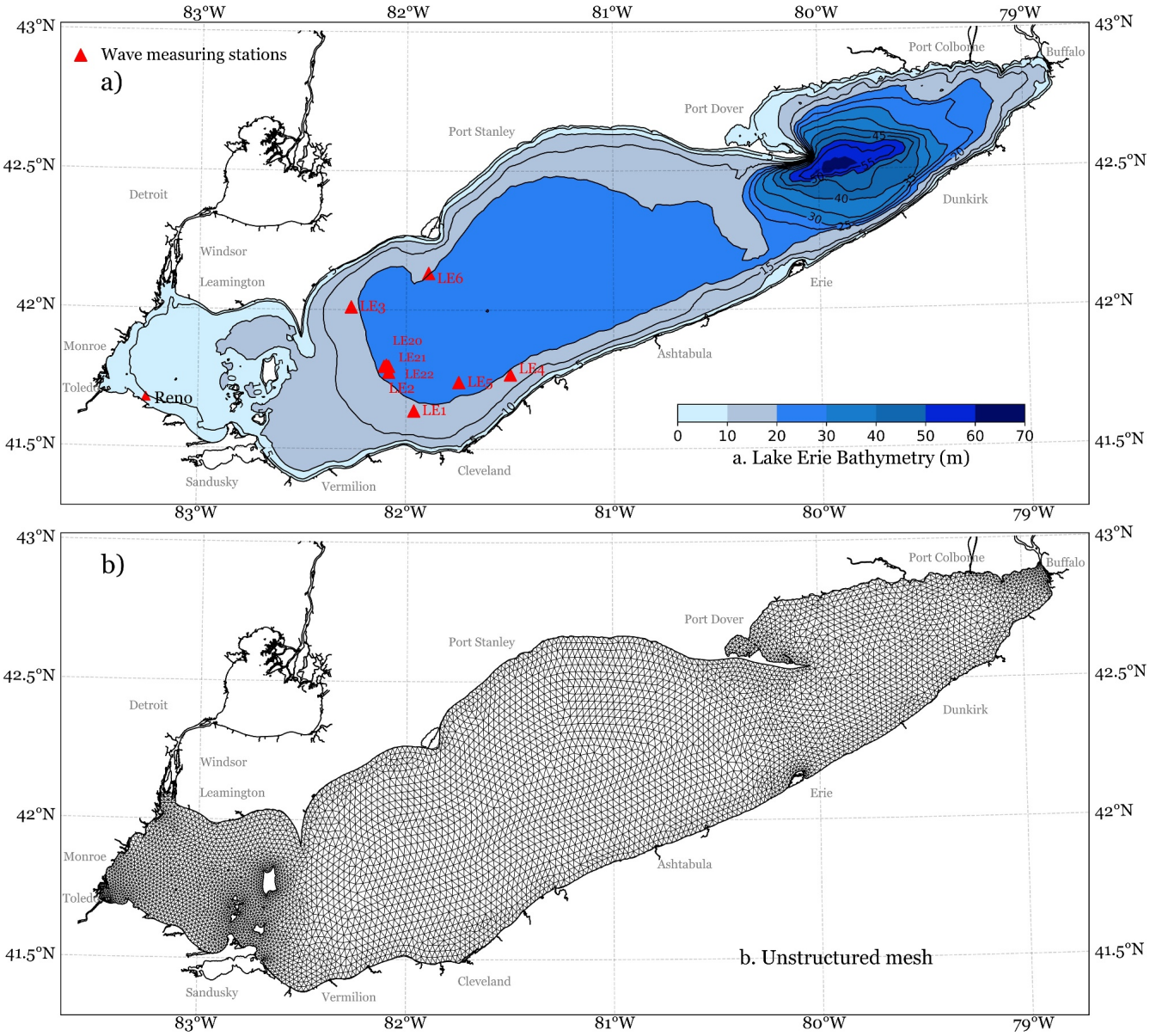
For ML research, the paper identifies a regional distribution-shift problem: polar spectral relationships need not be appropriate for shallow-lake spectra. A surrogate should be evaluated across seasons, stations, partial-ice conditions, and strong events, with held-out observations where possible. Such a design is a proposed follow-up, not an ML result in this paper.

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FIGURE ATLAS | 01 / 10

Figure 1. Bathymetry, stations, and model mesh



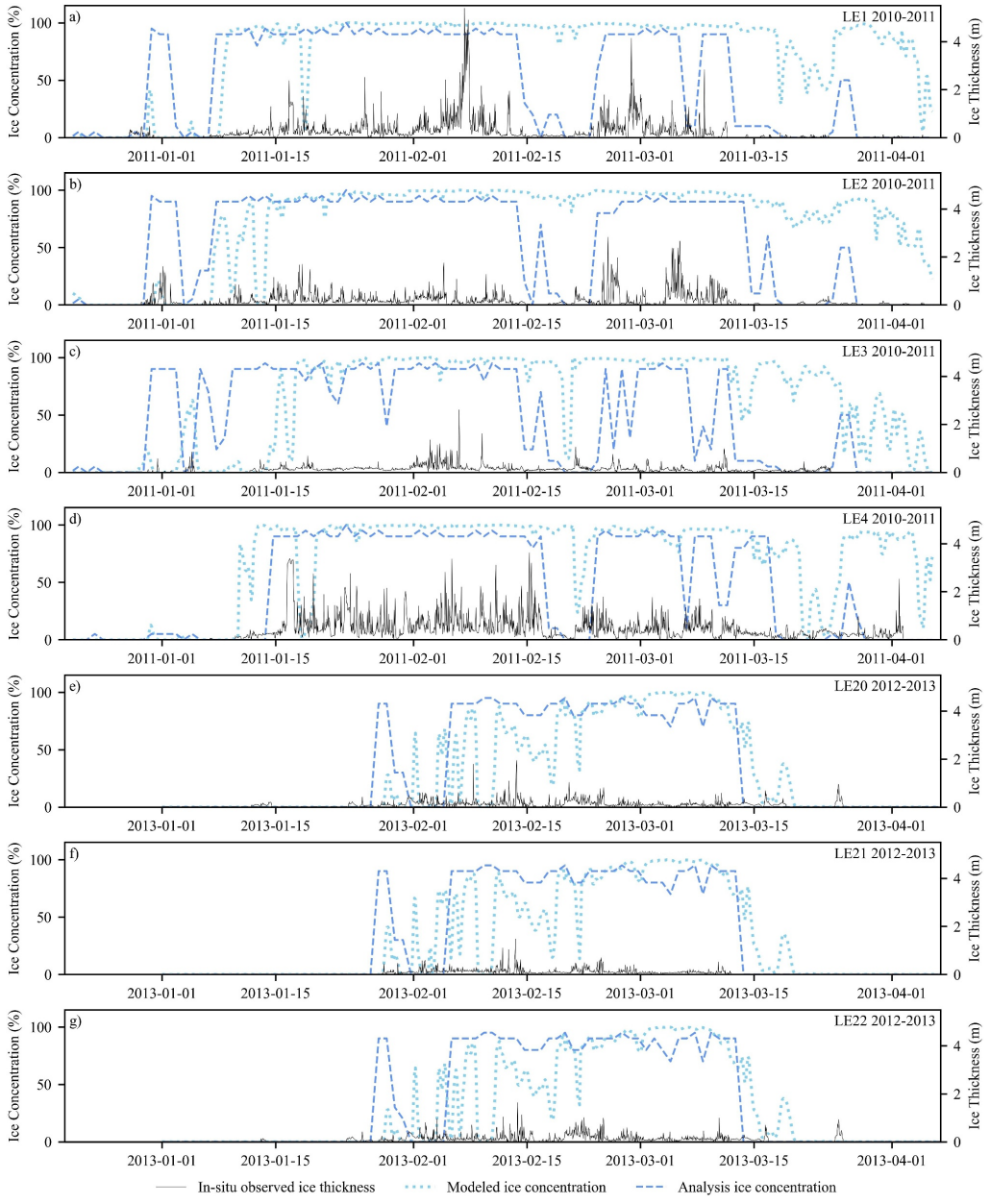
How to read this figure

Lake Erie bathymetry, in situ stations, and the unstructured mesh establish the shallow-water and spatial context of the experiments. Station coverage is concentrated in particular regions, so model skill at these sites should not automatically be generalized to every ice regime in the lake.

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FIGURE ATLAS | 02 / 10

Figure 2. Ice model, ice analysis, and sonar records



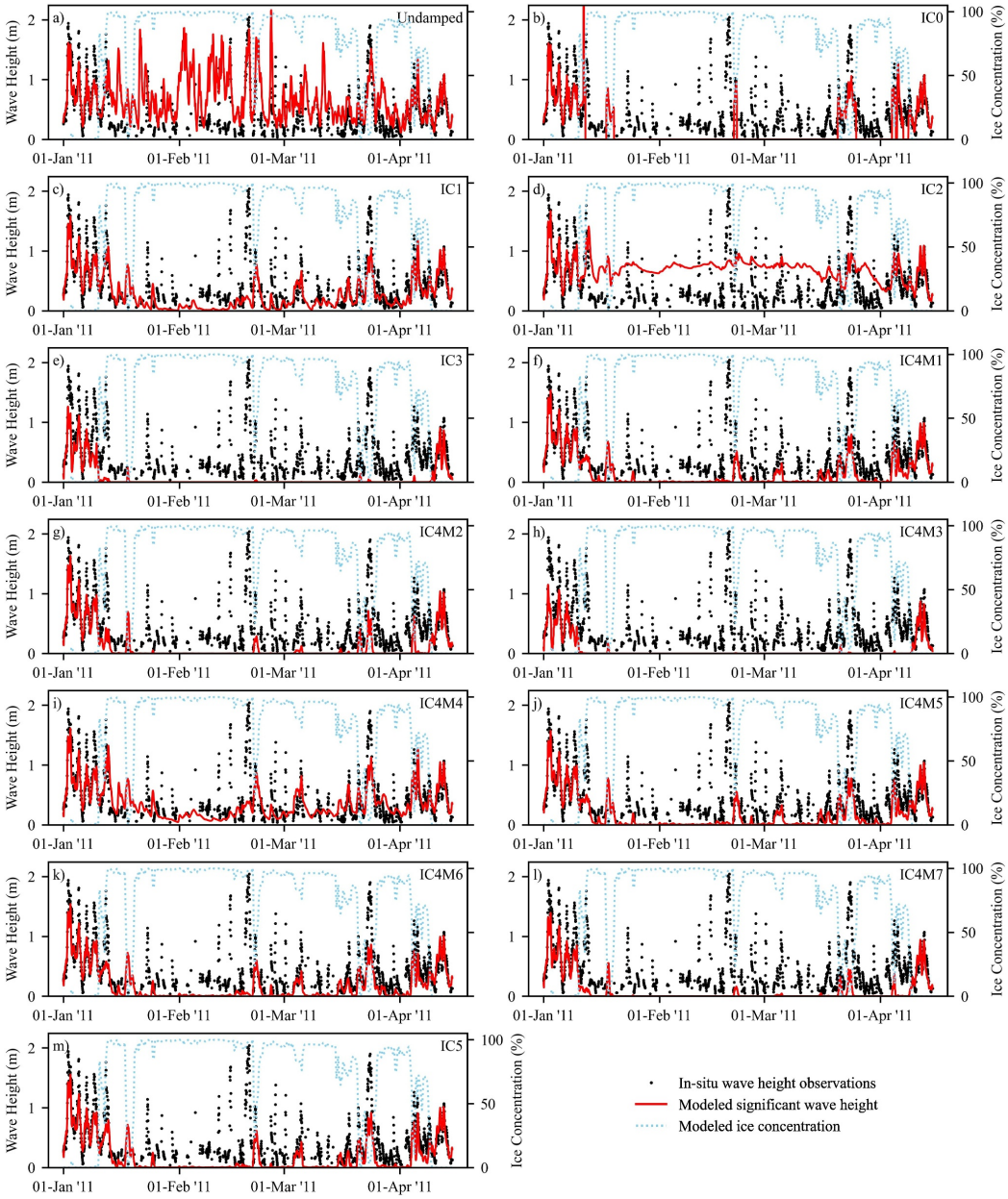
How to read this figure

The time series juxtapose point ice-thickness measurements with modeled and analyzed areal concentration. Differences near freeze-up, retreat, and episodic openings expose forcing uncertainty. These panels motivate the paper's presence/absence comparison while illustrating why the underlying observables are not identical.

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FIGURE ATLAS | 03 / 10

Figure 3. All tested damping configurations at LE4



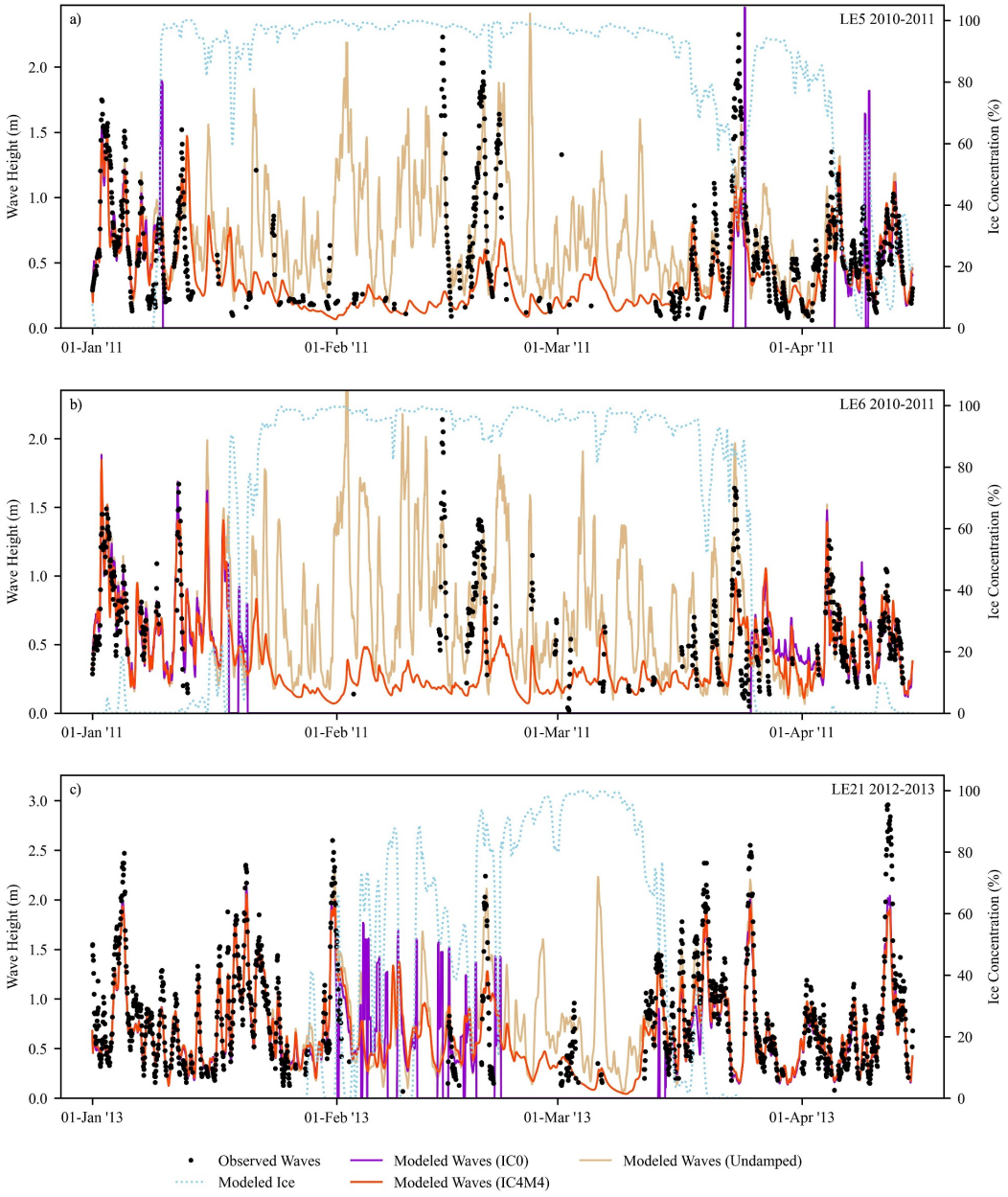
How to read this figure

The multi-panel comparison shows winter wave response for the twelve damping choices and the undamped reference. It reveals failure modes hidden by aggregate scores, including near-total suppression and insufficient event response. IC4M4 and IC1 preserve more of the observed variability than many of the strongly damped cases.

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FIGURE ATLAS | 04 / 10

Figure 4. Comparison across additional stations and winters



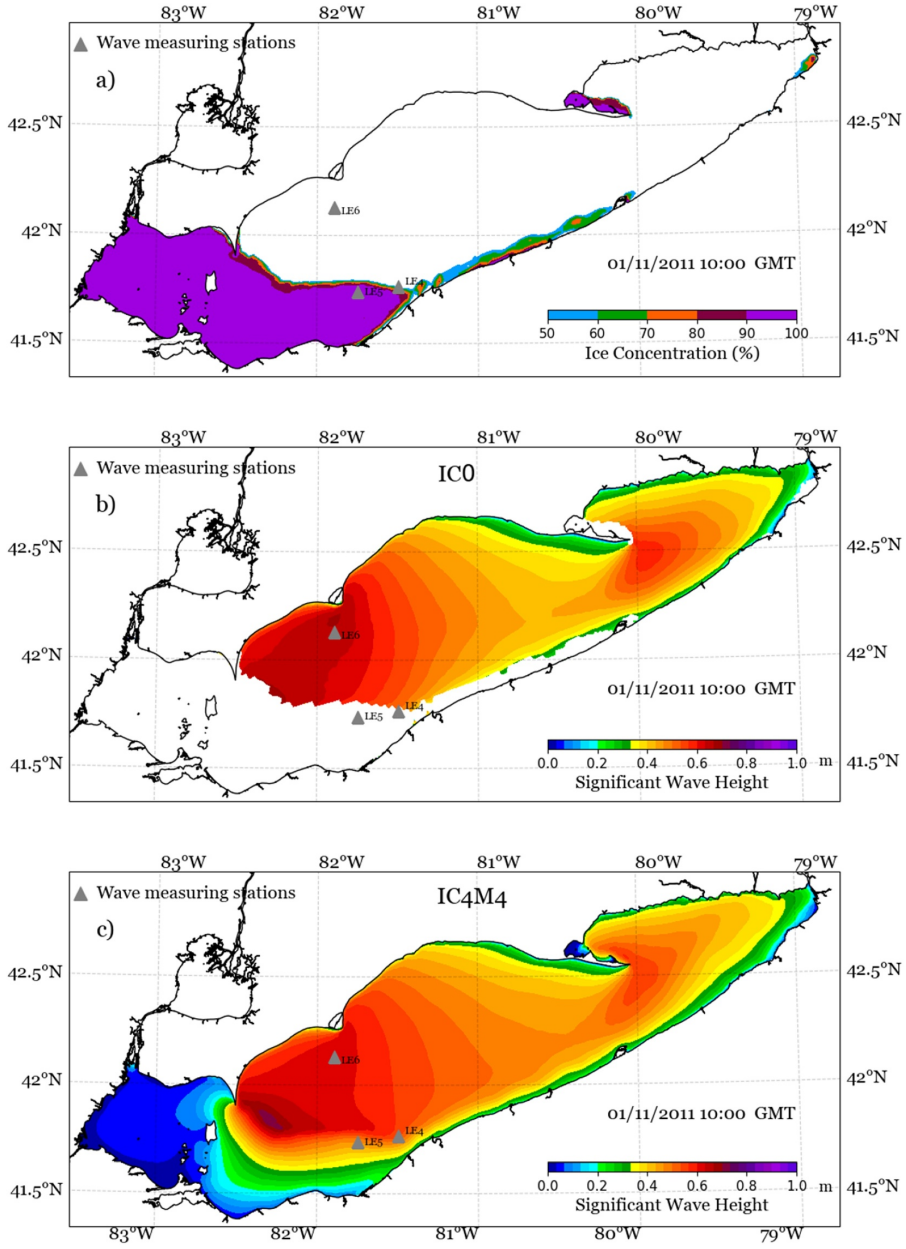
How to read this figure

The undamped, IC0, and IC4M4 simulations are compared with observations at LE5, LE6, and LE21. IC4M4 supplies wave information that IC0 removes, but still misses some energetic events. Variation in observation coverage must be considered when comparing performance between panels.

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FIGURE ATLAS | 05 / 10

Figure 5. Spatial effect of replacing the binary ice mask



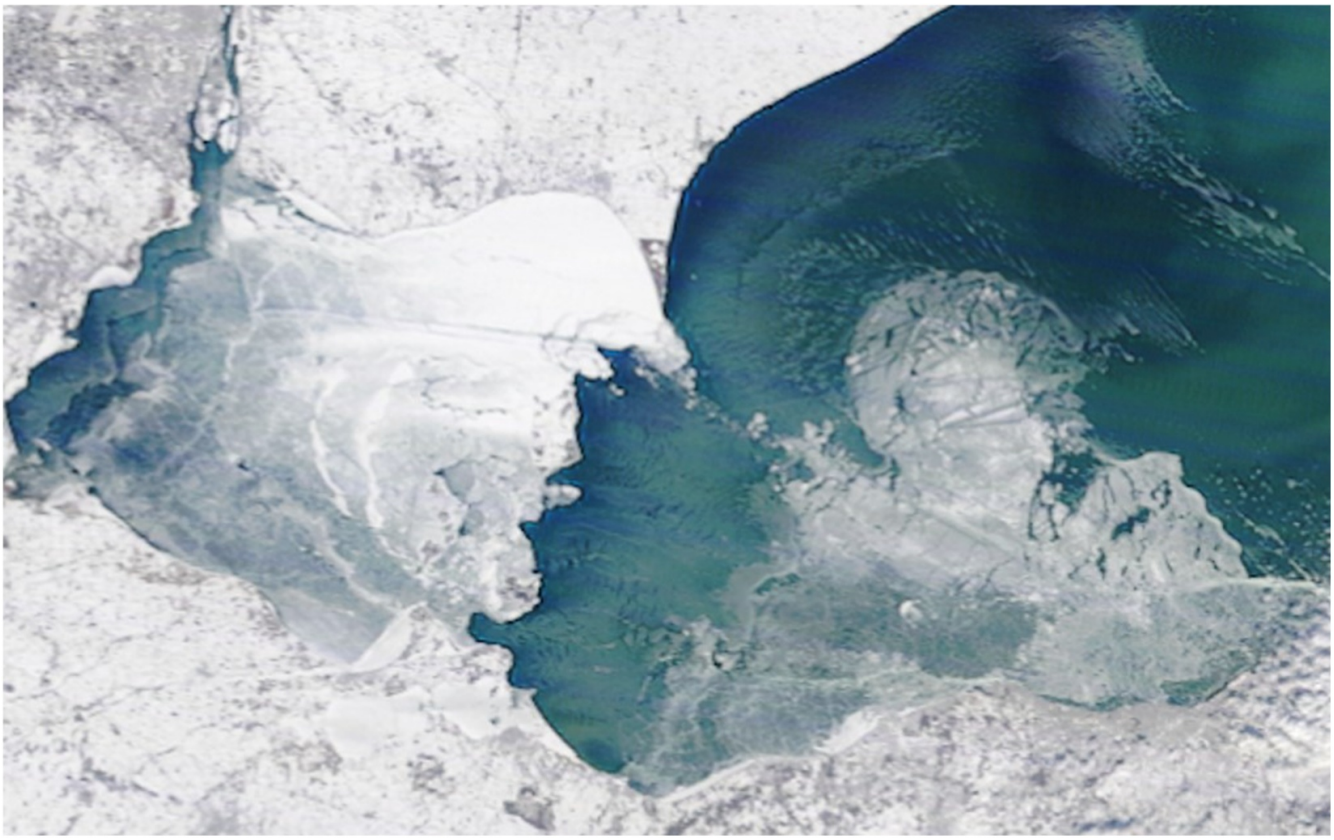
How to read this figure

The domain-wide snapshot compares the ice field and wave predictions under different ice treatments. A binary ice mask sharply truncates the forecast wave field; IC4M4 allows a gradual response within partial ice. The figure is evidence about spatial model behavior rather than a separate observational validation of every grid cell.

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FIGURE ATLAS | 06 / 10

Figure 6. Satellite view of the ice field



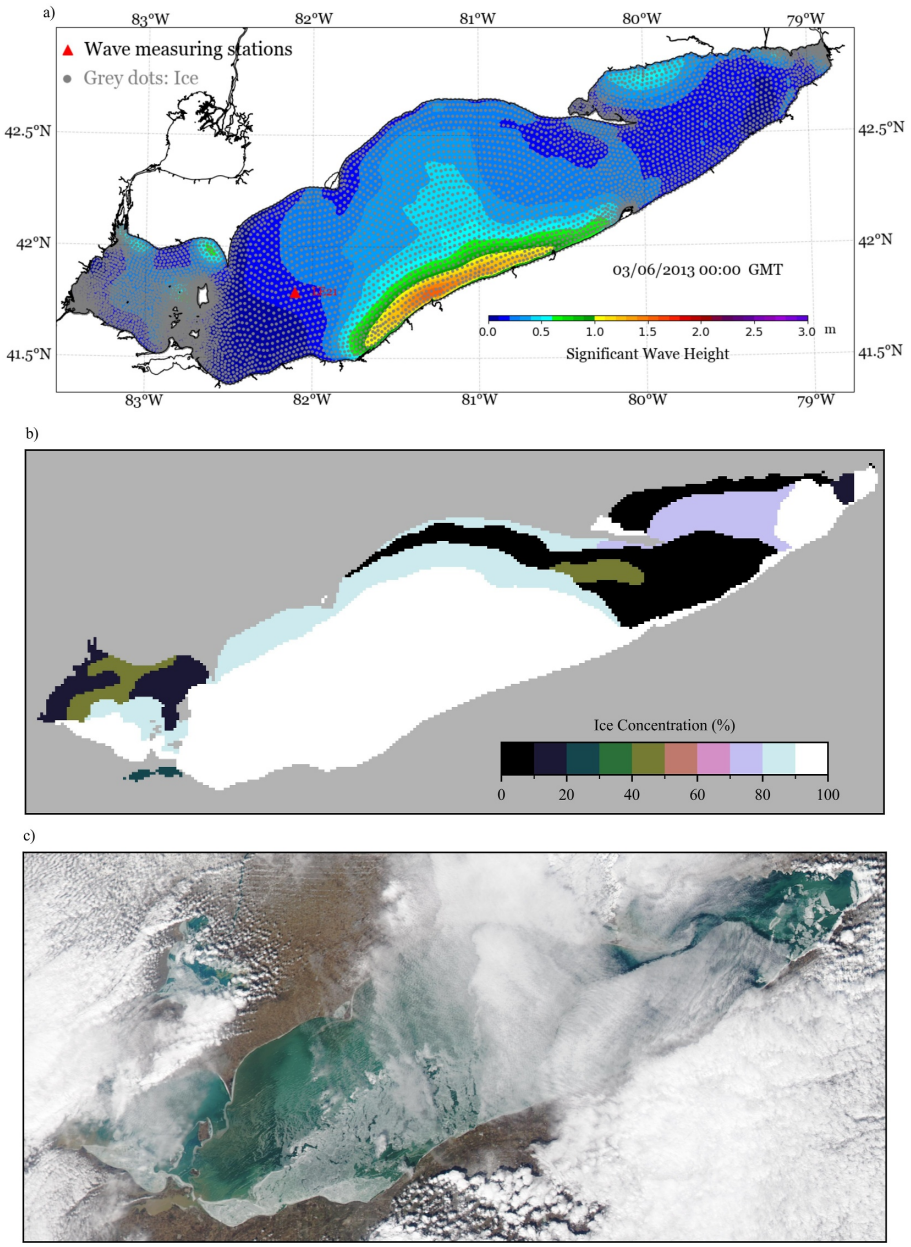
How to read this figure

The MODIS true-color scene provides independent qualitative context for the heterogeneous Lake Erie ice cover. Mixed ice and open-water features explain why a coarse all-or-nothing mask can be misleading. Cloud cover and interpretation limits still prevent treating imagery as perfect, instantaneous model truth.

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FIGURE ATLAS | 07 / 10

Figure 7. A second spatial case in March 2013



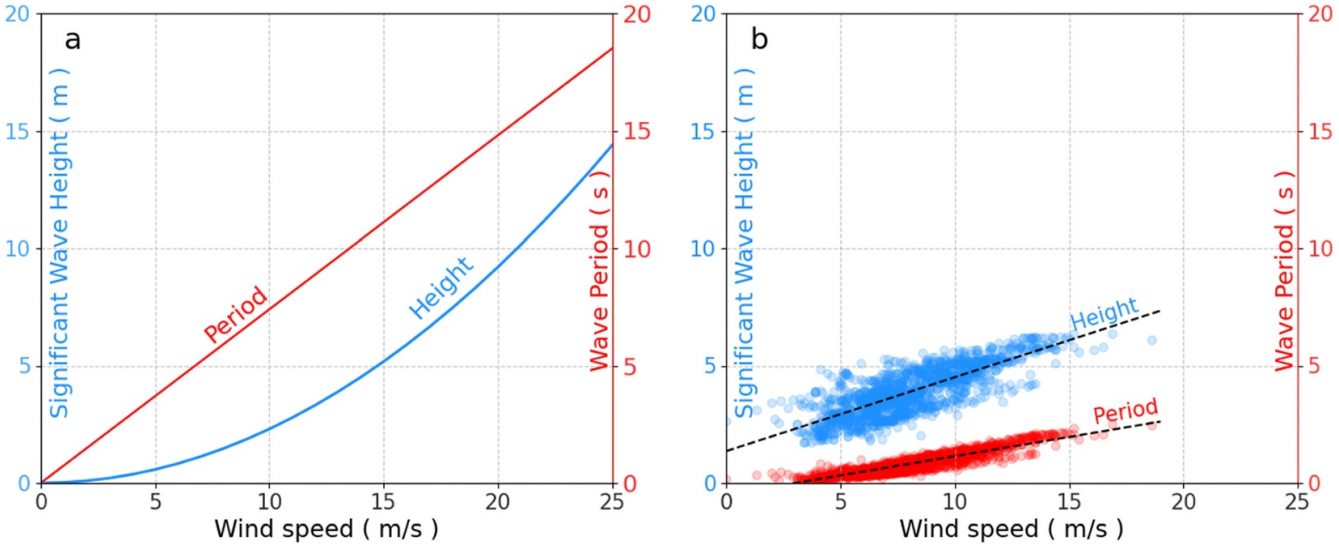
How to read this figure

Wave-height predictions are considered alongside simulated ice and satellite imagery during another winter event. The comparison connects local validation records to domain-scale ice structure and illustrates the persistence of the partial-ice forecasting problem outside the primary 2010-2011 comparison.

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Figure 8. Wave height and peak period versus wind



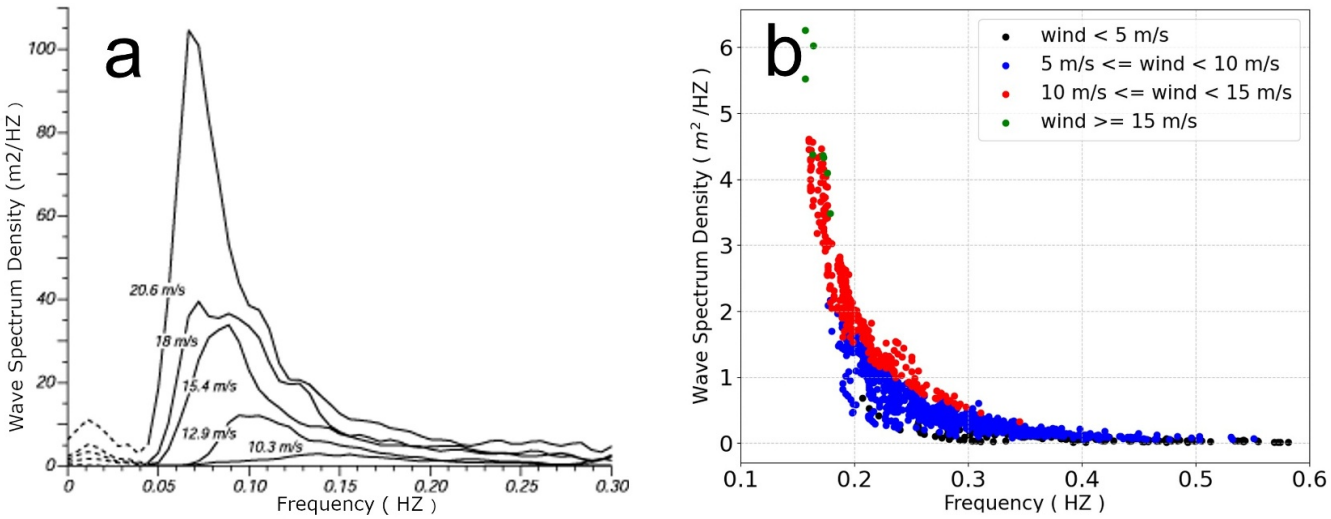
How to read this figure

The fully developed-sea reference and Lake Erie WW3 results show different wave-growth and peak-period relationships. Limited fetch and depth help distinguish lake waves from open-ocean conditions. These differences motivate checking the calibration range of a frequency-based ice attenuation law.

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Figure 9. Ocean and lake spectral shapes



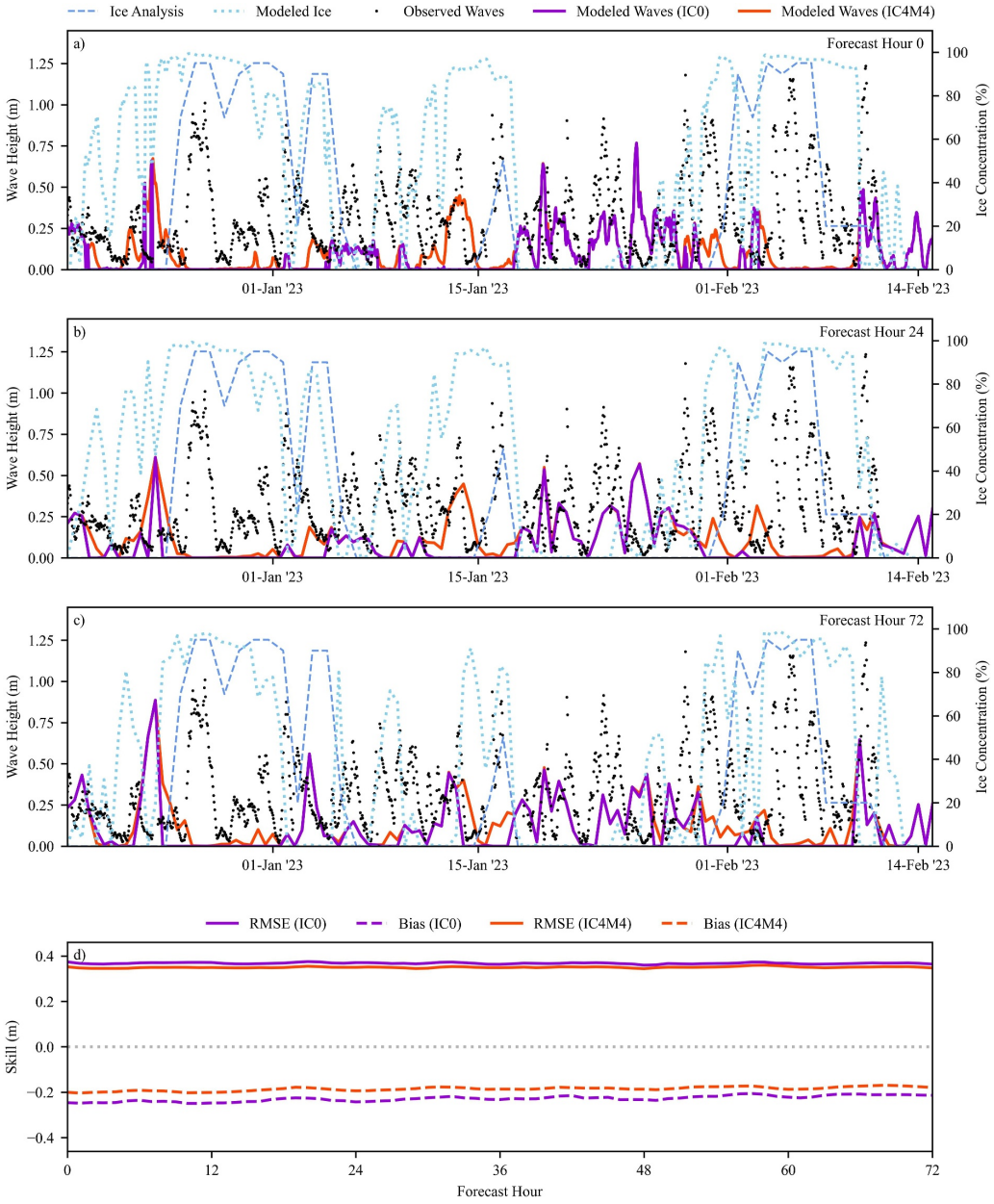
How to read this figure

Reference sea spectra and Lake Erie model spectra differ in peak frequency and response to wind speed. Spectral energy cannot be transferred between these settings by matching significant wave height alone. This is directly relevant to any conversion of lake-model spectra into a neural network's fixed spectral input bins.

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FIGURE ATLAS | 10 / 10

Figure 10. Pseudo-real-time forecast assessment



How to read this figure

Reno Beach observations are compared with IC0 and IC4M4 at several forecast lead times, together with RMSE and bias by lead time. The result extends the hindcast comparison to a forecast setting, while retained underprediction points to remaining uncertainty in ice fields and winter wind forcing.

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